KVIS HPC SHOWCASE · HPC & BRAIN CAVE

AI Research for Thai Stocks with HPC

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PROJECT IN BRIEF

The AI practises decisions and learns from feedback

Can a computer learn portfolio decisions from repeated practice instead of one fixed rule? The project compares PPO, A2C, DDPG, TD3 and SAC on Thai-stock data using one controlled process that measures return, risk and trading costs.

503
planned experiment runs

5
AI learning methods

29.7%
verified official-plan progress

WHERE HPC HELPS

HPC makes large experiments faster and checkable

A laptop is enough for coding and small tests. HPC schedules repeated runs, records job IDs, runtime and resources, and preserves evidence so the work can be checked or repeated.

CODE → SCHEDULE → RUN → VERIFY

SOFTWARE · Python · Stable-Baselines3 · Gymnasium · Optuna · SLURM

FROM CODE TO CHECKABLE EVIDENCE

1 WRITE THE RESEARCH CODE

One controlled process for five methods

```
for method in METHODS:
    train(method)
    verify(manifest, receipt)
```

2 WRITE THE HPC JOB SCRIPT

Request only the resources the job needs

```
#SBATCH --cpus-per-task=2
#SBATCH --mem=8GB
#SBATCH --time=02:00:00
#SBATCH --partition=compute-devel
```

3

ADD TEAM PHOTO

3 STUDENT RESEARCHERS

3 RUN AND MONITOR THE JOB

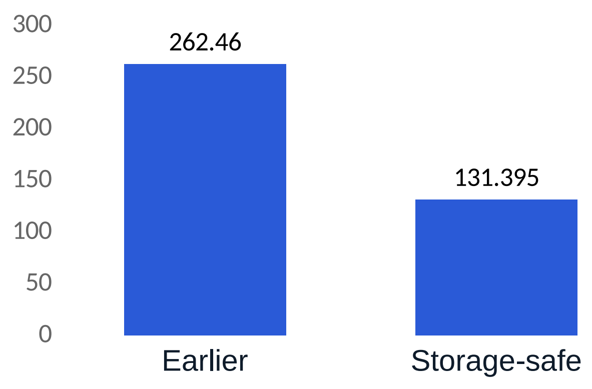
Every run has an identity, runtime and status

JOB 3550 **COMPLETED**

00:33:20 · 5 AI methods
2 CPUs · 809,680 KB peak RAM

4 CHECK THE RESULT

Less output without losing essential evidence



-49.94%

less test output
5 manifests + 5 receipts
0 checkpoints

WHY HPC MATTERS TO THE TEAM

N

NOTH

HPC removed the laptop bottleneck, but its biggest value was traceability. Every run keeps settings, a job ID, logs and measured resource use, so we can explain what happened.

MICKEY

The storage-safe workflow taught us that computing power needs control. We monitor storage, verify outputs and begin with a small safe test before scaling up.

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Verified: job 3550 completed · Next: recheck capacity before the 150-trial screening

Evidence through 17 Jul 2026 · Full screening remains unsubmitted · Research use only - not investment advice